Exploring soil management legacies in agronomy through EM and ERT

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Context and hypotheses

- **Persistent land-use and management legacies,** together with atmospheric forcing, control soil hydraulic behavior and thereby determine the duration of soil moisture anomalies from days to seasons (*Rahmati et al. 2024*).
- **Soil moisture is an Essential Climate Variable**, governing climate variability, hydrological extremes, agricultural irrigation, and sustainable water resource management.
- Understanding soil legacy is fundamental to explaining why soils respond differently to identical atmospheric forcing, shaping hydrological, ecological, and climate feedbacks (*Farmani et al. 2025*).

Methodology

- **Field site 1**: Experiment running for several growing seasons (from 2023) to investigate how soil moisture influences barley phenology and yield. The experimental is divided into four subplots, each receiving a different treatment: fertilization, irrigation, both, or neither (**fig 1 - left**).
- **Field site 2**: Long-term irrigation with saline water management in an irrigated vineyard under semi-arid conditions located in the Madrid region (central Spain) (**fig 1 - right**).

Results

- **Figure 2**: Surveys revealed **unexpected long-term soil legacy effects**: despite minimal topography and a shallow water table (<1 m), post-rainfall conditions were not uniform. EM and ERT measurements showed treatment-dependent EC differences near the surface and at 1 m depth.
- **Figure 3:** Uncertainties persisted in interpreting higher ER areas, as it was **unclear whether they stemmed from increased soil moisture or were linked to soil salinity caused by soil sodicity**.
- EM/ERT senses not just water content, but how water is distributed, how connected pores are, and solute pathways.

Perspectives

- Geophysical observations findings, highlight the importance of **exercising caution when performing initial measurements in an agronomic system,** as prior management and environmental history can strongly influence soil properties.
- Potential legacy effects can be indicated by the **difference between predicted and actual values:** further advances in land surface models and closer integration of model simulations (e.g. water balance mode, SaltMod/SahysMod for long-term salinisation trends) and observations are needed to better characterize moisture memory.
- Extend the concept to the carbon fluxes and ecological memory (*Canarini et al. 2021, Heinrich et al. 2025*).

Hypothesis: Long-term soil management practices (e.g., irrigation, fertilization) create persistent differences in soil structure and moisture retention, such that even after a saturating rainfall event, soil moisture and electrical conductivity (EC) vary among treatments.

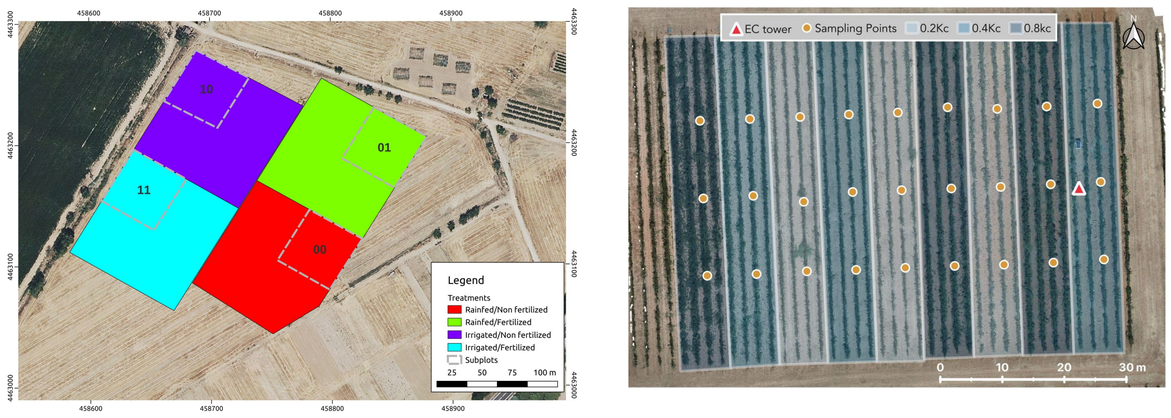

Figure 1: (left) La Poveda experimental site located near Madrid is a barley field (fallow during the survey) with four distinct treatments (fertilization, irrigation, both, or neither.) (right) El Socorro field site located South Madrid is a vineyard field with 3 differents irrigation regimes (0.2Kc under-irrigated, 0.4Kc control, 0.8Kc over-irrigated)

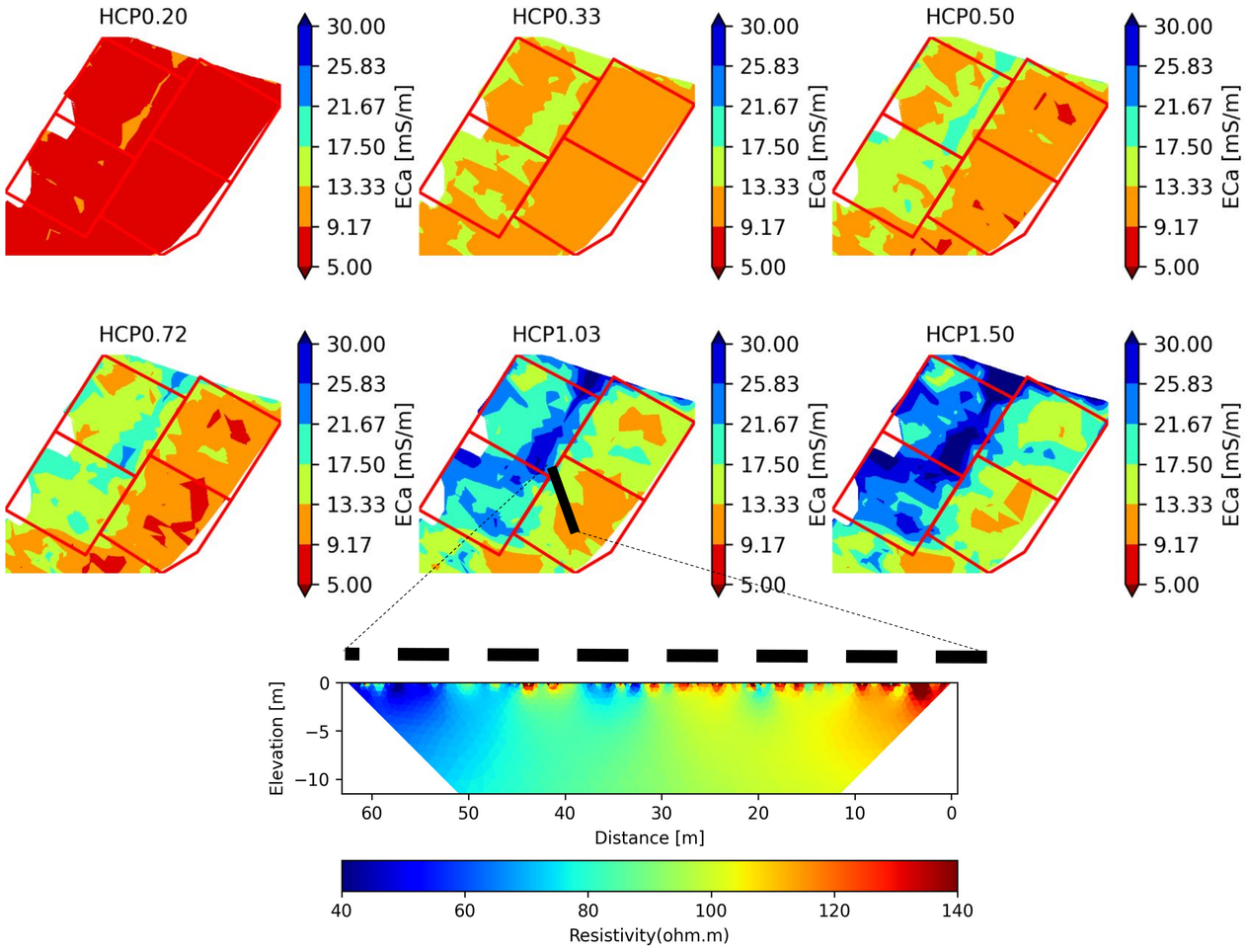

Figure 2: La Poveda field site in Septembre 2025 (top) EM survey apparent EC with a 6L mini-explorer (gf-instrument) conducted in high mode sensitivity (HCP - 0.20 is the coil distance) (bottom) ERT survey inverted resistivity section (4 roll-along with 0.5m electrodes inter-spacing)

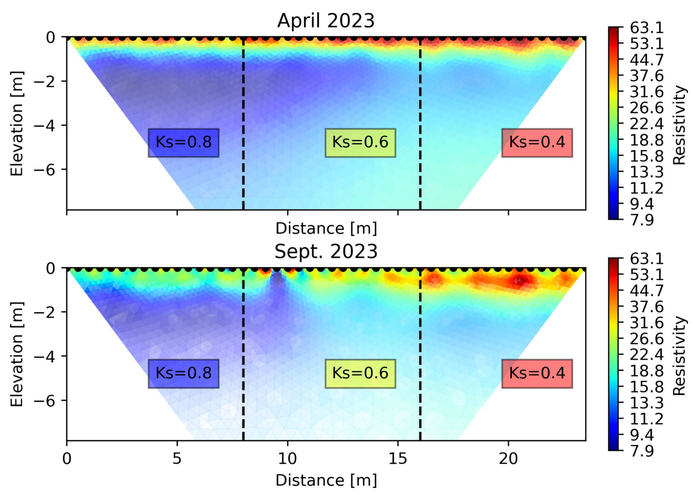

Figure 3: ER inverted model (Ω.m) in April and September 2023. The transect is 24m length (with 0.5m electrode spacing) intersect 3 differents irrigation treatments.

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ACKNOWLEDGEMENTS

This research was supported by the TWISTT project under the PRIMA Programme, funded by the European Union, and by the research project PCI2025-163228, financed by MICIU/AEI/10.13039/501100011033 and co-financed by the European Union. Benjamin Mary benefits from the grant "RyC2023-045040-I", funded by MICIU/AEI (grant no. 10.13039/501100011033) and the FSE. Field experiments in La Poveda were supported by EO4WUE project (TED2021-129814B-I00) funded by MCIN/AEI (DOI:10.13039/501100011033) and the European Union NextGenerationEU/PRTR. El Socorro experiments were supported by DATI project (PCI2021-121932) funded by MCIN/AEI (DOI:10.13039/501100011033) and the PRIMA EU program.